

02_analysis

Helix OTA — Exhaustive Requirement Analysis of additions/ initial_research_02.md

Field	Value
Revision	1
Created	2026-06-08
Last modified	2026-06-08
Status	active
Status summary	Line-by-line requirement inventory of the operator-supplied draft additions/ initial_research_02.md (~2044 lines), mapping every extractable requirement to the current canonical corpus (00-master/, 1.0.0-mvp/, 1.0.1-staged-rollout/, 1.X-*, research/adr/, research/stacks/) and the built Go server (server/). Each requirement is rated COVERED / PARTIAL / MISSING with a citation or gap, checked for conflict with the locked decisions/ADRs, and given a folding action. This is the granular successor to research/ additions_synthesis.md (Rev 1, high-level) for draft 02 specifically. HelixConstitution clause numbers (§11.4.x, §1, §7.1) are carried from corpus convention and are UNVERIFIED against the authoritative Constitution text (the Constitution file is not present in this repository). Several draft-02 factual specifics (hawkBit GitHub stats, “Mender client is C++”, Android-15 optional payload_properties header set) remain UNVERIFIED and are flagged
Issues	

Field	Value
	where they appear. Exact public surfaces of catalogue submodules (auth, security, Storage, database, http3, ratelimiter, middleware) are UNVERIFIED.
Fixed	Initial granular analysis of draft 02.
Fixed summary	All 15 numbered sections of draft 02 inventoried; coverage cross-referenced against corpus + built server code. Re-run if initial_research_02.md is revised. Feed the MISSING/PARTIAL items flagged below into the relevant phase specs (1.0.1 rollout, 1.X future-OS, ADR-0001 hawkBit un-gating). Confirm catalogue-brick public surfaces to clear the UNVERIFIED reuse claims.
Continuation	
Owner	Helix OTA spec-analysis
Related	../ additions_synthesis.md; ../../00-master/2026-06-07-helix-ota-design.md; ../adr/adr-0001-wrapped-engine.md ... adr-0005; ../../1.0.0-mvp/

Table of contents

1. purpose_and_method
2. coverage_summary
3. conflicts_with_locked_decisions
4. requirement_inventory
 - 4.1 research_existing_systems
 - 4.2 android15_ota_mechanism
 - 4.3 architecture
 - 4.4 database_schema
 - 4.5 api_specifications
 - 4.6 phased_rollout
 - 4.7 security_model
 - 4.8 testing_strategy
 - 4.9 infrastructure_deployment
 - 4.10 android_client
 - 4.11 future_os_support
 - 4.12 roadmap
 - 4.13 reusable_components
 - 4.14 success_factors
5. unverified_register
6. sources

1. purpose_and_method

Per locked decision D5 (additions/ is authoritative input to be deeply analyzed and folded in), this document inventories every requirement in additions/initial_research_02.md and maps it to the current state of the system. Three judgements are recorded per requirement:

- **Coverage** — COVERED (a corpus file or built code already realizes it, cited), PARTIAL (some but not all of it is realized; gap named), or MISSING (no corpus/code realization).
- **Conflict** — whether it contradicts a locked decision or an ADR; resolution recommended as *defer to ADR* / *reject* / *accept-with-modification* / *none*.
- **Folding action** — where the accepted element lands.

Evidence rule (§11.4.6 no-guessing / §7.1 anti-bluff, UNVERIFIED clause numbers): every COVERED rating cites a real file; unconfirmed claims are marked UNVERIFIED, never invented. Draft 02's self-description ("simulated analysis", "most comprehensive ever created") is marketing and is stripped, not folded (additions_synthesis.md §6).

Key finding up front: **the corpus is already mature and has folded the large majority of draft 02**. The master design, the five ADRs, the full 1.0.0-mvp/ spec set, and a working Go server (server/internal/api, server/internal/store) already realize draft 02's MVP-relevant content, *re-based onto the locked stack*. The residual gaps are concentrated in the **post-MVP** material (staged-rollout engine internals, TUF device client, future-OS adapters) which is correctly **deferred** (outlined, not specified) — plus a handful of draft-02 items that were deliberately **rejected** as conflicting with locked decisions (microservices, gRPC-primary, hawkBit-committed, TUF-in-MVP).

2. coverage_summary

Draft 02 section	Dominant rating	Notes
§1 research (hawkBit/Mender/ TUF/RAUC/ SWUpdate)	COVERED	Expanded into research/stacks/ * + ADR-0001/0002; draft's verdicts treated as hypotheses only.
§2 Android 15 Virtual A/B + payload.bin	COVERED	research/stacks/android15- virtual-ab.md, aosp-update- engine.md, 1.0.0-mvp/ client_android/ update_engine_integration.md.
§3 architecture (microservices)	PARTIAL / CONFLICT	Architecture COVERED as modular monolith (ADR-0003); microservices topology REJECTED for MVP.
§4 database schema	COVERED	1.0.0-mvp/database/schema.md + executed migration; richer-base normalization done.

Draft 02 section	Dominant rating	Notes
§5 REST + gRPC API	PARTIAL / CONFLICT	REST COVERED + built; gRPC demoted to optional/internal (out of scope, ADR-0004).
§6 phased rollout	PARTIAL (deferred)	Outlined in 1.0.1-staged-rollout/README.md; full engine spec + SQL not yet written.
§7 security (TUF in MVP)	PARTIAL / CONFLICT	MVP = signing+SHA-256+AVB (COVERED, built validator); TUF-in-MVP REJECTED, deferred to 1.0.1 (ADR-0002).
§8 testing	COVERED	Replaced by four-layer + mutation model (master §13); draft's 80% floor superseded.
§9 infra (compose + k8s)	COVERED	1.0.0-mvp/deployment/overview.md + compose/k8s manifests, re-based on containers.
§10 Android client (Kotlin + service)	COVERED	1.0.0-mvp/client_android/integration_guide.md (KMP + WorkManager); reference Kotlin reworked.
§11 future OS (Linux/Windows/universal)	PARTIAL (deferred)	OS-adaptor seam COVERED in master §4; per-OS specs are research outlines (1.X-*).
§12 48-week roadmap	PARTIAL	Re-expressed as phase dirs; week-by-week schedule not adopted as canonical.
§13 reusable components	PARTIAL / CORRECTED	Real catalogue used (submodule_reuse_map.md); draft's guessed submodule names corrected.
§14 success factors (SLAs/quality gates)	PARTIAL	Captured as guarantees (master §1); numeric SLAs (99.9%, <100ms, 10k devices) UNVERIFIED/not bound.

3. conflicts_with_locked_decisions

Five draft-02 positions conflict with locked decisions/ADRs. All are already resolved in the corpus; recorded here for traceability with the recommended resolution.

#	Draft-02 position (source)	Conflicts with	Recommended resolution	Corpus disposition
K1	Microservices topology with	D6/locked modular-		ADR-0003 decides modular mono is one binary. RESOLVED.

#	Draft-02 position (source)	Conflicts with	Recommended resolution	Corpus disposition
	7 services + API gateway (§3.2, L209–296)	monolith-for-MVP; ADR-0003	defer to ADR-0003 (reject for MVP; seams stay extractable)	
K2	gRPC service as a co-equal/primary device surface (§5.2, L557–624)	D6 REST-primary; ADR-0004 §4 (C4)	accept-with-modification: gRPC optional/internal only	endpoints.md §1 declares gRPC not built. RESOLVED.
K3	TUF in the MVP security service (§3.2.7, §7.1, L287–294, L758–833)	Locked scope (TUF→1.0.1+); ADR-0002 (C5)	defer to ADR-0002 (MVP = signing+SHA-256+AVB)	ADR-0002 defers device-side TUF validator built. RESOLVED.
K4	hawkBit committed as the deployment back end (§1.1, §3.2.4, L24–44, L258–266)	D3 engine choice is research-decided; ADR-0001 (C1)	defer to ADR-0001 (hawkBit GATED front-runner, AOSP-native fallback)	ADR-0001 keeps hawkBit gated; columns in MVP schema. RESOLVED.
K5	Early user/multi-version rollback roadmapped (§12 Phase 1.4.0, L1840–1858)	Locked MVP non-goal (end-user rollback deferred)	defer: MVP = automatic A/B boot-failure rollback only	master §1 non-goals; device_deployments.status= covers auto path. RESOLVED.

Secondary conflicts (non-architectural, already reconciled in additions_synthesis.md §5): 6 h poll cadence (§10.2 L1488) vs locked **15 min + jitter** (master D7) — **reject** draft constant; Redis as session store (§3.2.2/§9.1) — **accept-with-modification** (optional, prefer cache brick); 80% coverage floor (§8.1 L860) — **reject**, superseded by four-layer + mutation with $\geq 90\%$ on safety-critical paths (synthesis §5 C8).

4. requirement_inventory

ReqID format R02-<section>.<n>. Citations: corpus paths are relative to docs/research/main_specs/; server paths are relative to server/.

4.1 research_existing_systems (draft §1)

ReqID	Requirement (source)	Category	Coverage	Conflict?
R02-1.1	Eclipse hawkBit evaluated as wrap candidate (§1.1 L24)	Research	COVERED — research/ stacks/ eclipse- hawkbit.md; research/ adr/ adr-0001- wrapped- engine.md §3.1	K4 (hawkBit gated, not committed)
R02-1.2	hawkBit GitHub stats “579 stars... last commit June 5 2026” (§1.1 L28)	Research	MISSING / UNVERIFIED — not propagated as fact anywhere (synthesis §6 flagged)	none
R02-1.3	Mender evaluated; “C++ client, not Go” (§1.2 L46–60)	Research	COVERED — research/ stacks/ mender.md; ADR-0001 §3.2. “C++ client” UNVERIFIED (synthesis §6)	K4/C1
R02-1.4	TUF (go-tuf/v2) as security framework (§1.3 L62)	Research	COVERED — research/ stacks/tuf- go-tuf.md; ADR-0002 §3.2	K3 (TUF→1.0.1)
R02-1.5	RAUC for Linux (§1.4 L80)	Research	COVERED — research/ stacks/ rauc.md; 1.X-linux/ README.md §2	none
R02-1.6	SWUpdate for embedded Linux (§1.5 L91)	Research	COVERED — research/ stacks/ swupdate.md; 1.X-linux/ README.md §2	none
R02-1.7		Architecture	COVERED as gated option	K4

ReqID	Requirement (source)	Category	Coverage	Conflict?
	Wrap-hawkBit integration topology (Go→REST→hawkBit→DDI→devices) (§1.1 L38)		— ADR-0001 §3.1; landscape report §3.2	

4.2 android15_ota_mechanism (draft §2)

ReqID	Requirement (source)	Category	Coverage
R02-2.1	Virtual A/B with compression (Android 15) (§2.1 L106)	Client	COVERED — stacks/ab.md; n
R02-2.2	10-step update flow poll→download→verify→applyPayload→reboot→verify→rollback (§2.2 L115)	Client	COVERED — client_integrat cycle; ma
R02-2.3	OTA .zip structure (payload.bin, payload_properties.txt, META-INF, caremap) (§2.3 L130)	Client	COVERED — client_update_
R02-2.4	payload.bin internal structure (metadata/manifest/signatures/blobs) (§2.4 L144)	Client	COVERED — stacks/engine. update_e
R02-2.5	update_engine.applyPayload(url, offset, size, props) API + callbacks (§2.5 L161)	Client	COVERED — stacks/engine-endpoint offset/siz
R02-2.6	payload_properties FILE_HASH/FILE_SIZE/METADATA_HASH/ METADATA_SIZE (§10.1 L1329)	Client/ API	COVERED — §12.1 pa wire.go

4.3 architecture (draft §3)

ReqID	Requirement (source)	Category	Coverage	Conflict?	Folding action
R02-3.1	Universal OS-agnostic OTA with pluggable OS adapters (§3 L18, §11.3)	Architecture	COVERED — master §4 OS-adapter seam; 1.X-*/ README.md §3 adapter contract	none	None
R02-3.2	Client layer (Android/Linux/	Architecture	PARTIAL — Android COVERED (client_android/); Linux/	none	Deferred to future phases

ReqID	Requirement (source)	Category	Coverage	Conflict?	Folding action
	Windows clients) (§3.1 L199)		Windows are research outlines (1.X–linux, 1.X–windows)		
R02-3.3	API gateway (Traefik/Kong/custom) (§3.2.1 L227)	Architecture	PARTIAL — reverse proxy in deployment/overview.md §3.4; no separate gateway service (monolith)	K1	Accept reverse proxy only; reject gateway as-service
R02-3.4	Auth service (JWT/OAuth2/RBAC) (§3.2.2 L237)	Architecture	COVERED — as a module: endpoints.md §4; server/internal/api/middleware.go requireRole, token.go, users.go	K1 (module not service)	None
R02-3.5	Artifact service (upload/store/validate/checksum) (§3.2.3 L247)	Architecture	COVERED — server/internal/api/handlers_artifact.go; 1.0.0-mvp/server/artifact_validation.md	K1	None
R02-3.6	Deployment service wrapping hawkBit (§3.2.4 L258)	Architecture	PARTIAL — deployment module built (handlers_deployment.go), all-targets only; hawkBit wrapped	K1, K4	Defer hawkBit to ADR-0001
R02-3.7	Device management service (registration/inventory/groups) (§3.2.5 L268)	Architecture	COVERED — server/internal/api/handlers_device.go; store.go Device	K1	Groups schema present, group endpoints deferred (see R02-5.5)
R02-3.8	Monitoring service (Prometheus/Grafana/ELK) (§3.2.6 L278)	Architecture	PARTIAL — telemetry ingest COVERED (handlers_client.go, server/telemetry_processing.md); Prometheus/Grafana/ELK wiring not built	K1	OpenTelemetry via observability brick (master §3)
R02-3.9	Security service (TUF/signing/key mgmt/HSM/Vault) (§3.2.7 L287)	Architecture	PARTIAL — signing/verify + key mgmt COVERED (security/signing_verification.md, key_management.md); TUF/HSM/Vault deferred	K3	Defer TUF/HSM to ADR-0002/1.0
R02-3.10	Data layer: PostgreSQL/Redis/MinIO/Prometheus (§3.1 L222)	Architecture	PARTIAL — PostgreSQL + MinIO COVERED (master §3, deployment/); Redis optional (cache brick); Prometheus surface only	C7 (Redis)	Accept PG+MinIO; Redis only if needed

4.4 database_schema (draft §4)

All of draft §4 is COVERED by 1.0.0-mvp/database/schema.md (12 tables, executed migration on PG 16.14). Draft 02 was explicitly adopted as the **richer base**, normalized (master §7; schema.md §1).

ReqID	Requirement (source)	Category	Coverage	Conflict?	Folding action
R02-4.1	users + api_keys (§4.1 L304–323)	Schema	COVERED — schema.md §5.1–5.2	none	None
R02-4.2	devices + device_groups + device_group_members (§4.1 L326–355)	Schema	COVERED — schema.md §5.3–5.4	none	None
R02-4.3	artifacts + artifact_versions (§4.1 L358–385)	Schema	COVERED — schema.md §5.5 (checksum_sha256 NOT NULL + CHECK; signature col)	none	None
R02-4.4	deployments (§4.1 L388–401)	Schema	COVERED — schema.md §5.7 (rollout_strategy JSONB; all_at_once default)	none	None
R02-4.5	deployment_phases (§4.1 L403–414)	Schema	MISSING (deliberately) — deferred to 1.0.1 migration 002_*	none	schema.md §8; 1.0.1–staged-rollout §49
R02-4.6	device_deployments (§4.1 L416–427)	Schema	COVERED — schema.md §5.8 (phase_id omitted at MVP)	none	None
R02-4.7	rollouts incl. hawkbit_rollout_id (§4.1 L430–441)	Schema	MISSING (deliberately) — hawkBit gated; deferred	K4	schema.md §8; un-gate with ADR-0001
R02-4.8	audit_logs (§4.1 L444–454)	Schema	COVERED — schema.md §5.10	none	None
R02-4.9	update_metrics (§4.1 L457–469)	Schema	COVERED — renamed to canonical telemetry_events, schema.md §5.9	none	None (rename)
R02-4.10	Performance indexes (§4.1 L472–483)	Schema	COVERED — schema.md §7 (45 indexes, draft list +	none	None

ReqID	Requirement (source)	Category	Coverage	Conflict?	Folding action
			query-driven additions)		

4.5 api_specifications (draft §5)

ReqID	Requirement (source)	Category	Coverage	Conflict?	Folding action
R02-5.1	REST /api/v1 base + auth/artifacts/deployments/devices endpoints (§5.1 L490–555)	API	COVERED — endpoints.md; built routes in server/internal/api/server.go (auth, devices, artifacts, releases, deployments, client)	none	None
R02-5.2	gRPC UpdateService (CheckForUpdate/Download/ReportStatus/RequestRollback) (§5.2 L557–624)	API	MISSING (deliberately) — gRPC out of scope; REST equivalents exist	K2	Reject as primary; optional/internal only
R02-5.3	Auth endpoints login/logout/refresh/device-register (§5.1 L493–498)	API	PARTIAL — login/refresh/register COVERED (server.go L112–119); explicit / logout not present (refresh-rotation model)	none	Accept-with-mod: no stateful logout in MVP
R02-5.4	Artifact CRUD + validate + download (§5.1 L501–509)	API	PARTIAL — upload/get COVERED; PUT/DELETE/explicit validate/download-route not built (validate is inline on upload; download is the	none	Accept-with-mod; CRUD-delete deferred

ReqID	Requirement (source)	Category	Coverage	Conflict?	Folding action
R02-5.5	Deployment lifecycle start/pause/resume/rollback + status/devices (§5.1 L512–524)	API	Range-served storage path) PARTIAL — create/get COVERED; start/pause/resume/rollback/devices-list are rollout-engine ops deferred to 1.0.1	K5 (rollback)	Deferred to 1.0.1–staged-rollout
R02-5.6	Device CRUD + history + check-update (§5.1 L527–535)	API	PARTIAL — register/status/check-update COVERED; device PUT/DELETE/history not built	none	Accept-with-mod; history deferred
R02-5.7	Device-groups CRUD + membership (§5.1 L538–546)	API	MISSING — group tables exist (schema §5.4) but no group endpoints built	none	Deferred; needed for grouped deployments (1.0.1)
R02-5.8	Monitoring endpoints (dashboard/metrics/logs/alerts) (§5.1 L549–555)	API	MISSING — telemetry ingest built; read-side dashboard/metrics/alerts endpoints not built	none	Deferred to dashboard/monitoring phase

4.6 phased_rollout (draft §6)

ReqID	Requirement (source)	Category	Coverage	Conflict?	Folding action
R02-6.1	Phased rollout JSON config (ordered phases: %, success/error thresholds, duration, auto_progress) (§6.1 L632–683)	Rollout	PARTIAL — design seed adopted: 1.0.1–staged-rollout/README.md §33; master §8. Full schema/SQL	none	Specify in 1.0.1–staged-rollout

ReqID	Requirement (source)	Category	Coverage	Conflict?	Folding action
R02-6.2	Phase names Canary/Pilot/Limited/GA (§6.1 L646–676)	Rollout	not yet written PARTIAL — pattern noted (Foundries wave/canary, 1.0.1 §33); not formalized	none	Specify in 1.0.1
R02-6.3	Rollout engine Go loop (start→monitor→threshold-check→advance/pause) (§6.2 L687–749)	Rollout	PARTIAL — adopted as design reference, hardened with deterministic cohort selection + idempotent transitions (master §8; 1.0.1 §33); not implemented	none	Implement ota-rollout-engine in 1.0.1
R02-6.4	pause_on_error / rollback_on_critical_failure / halt-on-breach (§6.1 L678–682)	Rollout	PARTIAL — “halt wins over advance” safety invariant captured (1.0.1 §33); auto-abort wiring is 1.0.1 (§37)	none	1.0.1
R02-6.5	Notification channels (email/slack) (§6.1 L681)	Rollout/Ops	MISSING — alerting routed to Herald brick (master §9) but channels not specified	none	Specify in 1.0.1 monitoring via Herald

4.7 security_model (draft §7)

ReqID	Requirement (source)	Category	Coverage	Conflict?	Folding action
R02-7.1	TUF metadata signing (Root/Targets/Snapshot/Timestamp) (§7.1 L758–798)	Security	MISSING (deliberately) — deferred; ADR-0002 §3.2 (device enforcement gated)	K3	1.0.1 device client
R02-7.2	Artifact verify: signature + SHA-256/512 hash match before apply (§7.1 L800–832)	Security	COVERED — security/signing_verification.md §5–6; server/internal/api/handlers_artifact.go validation; device re-verify (integration_guide.md §7)	none	None
R02-7.3	Auth flow: login→JWT→Redis session→validate-per-request (§7.2 L837–851)	Security	PARTIAL — OAuth2/JWT + refresh-rotation COVERED (endpoints.md §4; token.go); Redis session store optional (cache brick)	C7	Accept; Redis if needed
R02-7.4	Key management (root/targets/snapshot/timestamp keys, HSM, Vault) (§7.1 L765–770)	Security	PARTIAL — MVP signing-key custody COVERED (security/key_management.md); multi-role TUF keys + HSM/Vault deferred	K3	1.0.1 / ADR-
R02-7.5	Zero-trust architecture (§14.1 L1960)	Security	PARTIAL — TLS 1.3, per-request JWT verify, device-id binding (master §6); full zero-trust not formalized	none	Threat mode master/threat_mod

4.8 testing_strategy (draft §8)

ReqID	Requirement (source)	Category	Coverage	Conflict?	Folding action
R02-8.1	Test levels (unit/integration/e2e/load/security) (§8.1 L859–889)	Testing	COVERED — superseded by four-layer + mutation model (master §13; tests/test_strategy.md); built tests present (server/internal/api/*_test.go)	none	None
R02-8.2	80% coverage minimum (§8.1 L860)	Testing	COVERED-with-override — four-layer + mutation immunity, ≥90% floor on safety-	C8	Reject flat 80%; per-component floor

ReqID	Requirement (source)	Category	Coverage	Conflict?	Folding action
			critical paths (synthesis §5 C8)		
R02-8.3	Example tests (upload/phased-rollout/rollback) (§8.2 L893–977)	Testing	PARTIAL — upload/auth/device/deployment tests built (handlers_*.test.go); phased-rollout + rollback tests are 1.0.1	none	1.0.1 for rollout/rollback tests
R02-8.4	Load test 10,000+ concurrent devices (§8.1 L878, §14.1 L1958)	Testing	MISSING / UNVERIFIED — no load-test harness or figures; rate-limit numbers UNVERIFIED (endpoints.md §5)	none	Defer; set numbers from MVP load tests

4.9 infrastructure_deployment (draft §9)

ReqID	Requirement (source)	Category	Coverage	Conflict?	Folding action
R02-9.1	Docker Compose dev stack (§9.1 L986–1105)	Infra	COVERED — 1.0.0-mvp/deployment/overview.md + docker-compose.mvp.yml, re-based on containers brick; hawkBit service dropped (gated)	K4	None
R02-9.2	Kubernetes manifests (deployment/service/statefulset/probes) (§9.2 L1109–1240)	Infra	COVERED — deployment/overview.md §6 + kubernetes/manifests	none	None
R02-9.3	Image digest pinning / no :latest (implicit)	Infra	COVERED — overview.md §11 anti-bluff fixes (minio pinned, digest TBD)	none	None
R02-9.4	Secrets via env/Secret refs (§9 various)	Infra	COVERED — overview.md §5 secrets handling	none	None

4.10 android_client (draft §10)

ReqID	Requirement (source)	Category	Coverage
R02-10.1	HelixOTAClient lib (checkForUpdate/downloadAndInstall/rollback) (§10.1 L1248–1463)	Client	COVERED — reworked to KMP in 1.0.0-mvp/client_android/integration_guide.md (register→poll→download→verify→apply→report)
R02-10.2	Verify checksum before apply (§10.1 L1319)	Client	COVERED — integration_guide.md §7 verify-before-apply (Security-KMP)
R02-10.3	file:/// local apply path (§10.1 L1328)	Client	COVERED — local verified-file apply chosen (ADR-0002 §4.1; integration_guide.md §7); resolves draft's file:// vs HTTPS ambiguity (synthesis §6)
R02-10.4	Foreground OTAUpdateService, 6 h periodic check (§10.2 L1480–1524)	Client	PARTIAL — WorkManager PeriodicWorkRequest COVERED (integration_guide.md §6); 6 h cadence rejected for locked 15 min + jitter
R02-10.5	Notification channel + progress UI (§10.2 L1541–1575)	Client	PARTIAL — duty cycle covers status reporting; notification UI not specified in MVP spec
R02-10.6	Rollback via marking slot unbootable / root (§10.1 L1371–1381)	Client	COVERED-by-native — automatic A/B boot-failure rollback (AVB/boot_control), not app-driven (master §6; research/stacks/android-avb-rollback.md)

4.11 future_os_support (draft §11)

ReqID	Requirement (source)	Category	Coverage	Conflict?	Folding action
R02-11.1	Linux support via RAUC/SWUpdate/pkg-mgr adapters (§11.1 L1591–1626)	Future-OS	PARTIAL (deferred) — 1.X-linux/README.md research outline + adapter table; not specified in depth	none	Future phase
R02-11.2	Windows support (Windows Update API/MSI/MSIX/GroupPolicy) (§11.2 L1628–1655)	Future-OS	PARTIAL (deferred) — 1.X-windows/README.md outline (MSIX/MSI/WinGet/WUA)	none	Future phase

ReqID	Requirement (source)	Category	Coverage	Conflict?	Folding action
R02-11.3	Universal OSAadapter interface + AdapterRegistry (§11.3 L1657–1689)	Future-OS	COVERED (seam) — master §4 OS-adapter seam; 1.X-*/README.md §3 contract (CheckForUpdate/Download/Verify/Install/Rollback/GetCapabilities)	none	Seam defined; registry impl future
R02-11.4	RTOS / macOS support (§12 Phase 2.0.0 L1868–1872)	Future-OS	PARTIAL (deferred) — 1.X-other-os/README.md	none	Future phase

4.12 roadmap (draft §12)

ReqID	Requirement (source)	Category	Coverage	Conflict?	Folding action
R02-12.1	48-week, 8-phase roadmap (1.0.0... 2.0.0) (§12 L1693–1885)	Planning	PARTIAL — re-expressed as phase directories (master §11); week-by-week schedule NOT adopted as canonical	none	Keep phase dirs; drop calendar
R02-12.2	Phase 1.0.0 MVP (8 weeks) scope (§12 L1695–1726)	Planning	COVERED — master §5 MVP definition matches (auth/artifact/device/deploy/telemetry/android)	none	None
R02-12.3	Phase 1.0.1 phased rollout (§12 L1728–1747)	Planning	COVERED — 1.0.1-staged-rollout/ (note: corpus folds rollback+TUF into 1.0.1, draft split them later)	none	None

ReqID	Requirement (source)	Category	Coverage	Conflict?	Folding action
R02-12.4	Phases 1.0.2 monitoring / 1.1.0 TUF / 1.2.0 Linux / 1.3.0 Windows / 1.4.0 rollback / 2.0.0 universal	Planning	PARTIAL — mapped to phase dirs but renumbered/ regrouped vs draft	none	Corpus numbering wins

4.13 reusable_components (draft §13)

ReqID	Requirement (source)	Category	Coverage
R02-13.1	Reuse vasic-digital bricks: containers/ security/auth/ config/ observability/ cache/storage/ eventbus/recovery/ streaming (§13 L1892–1932)	Reuse	COVERED-with-correction — real catalogue in 00-mas (auth,security,database,Storage,observability,etc.)
R02-13.2	Reuse HelixDevelopment: HelixConstitution/ helixqa/HelixCode/ LLMProvider (§13 L1934–1951)	Reuse	PARTIAL — Constitution governs corpus (UNVERIFIED)
R02-13.3	Guessed submodules (go-common, helm-charts, vasic-digital/ secrets)	Reuse	CORRECTED — do not exist; replaced by verified catalog
R02-13.4	NEW submodules (ota-protocol, ota-artifact-validator, ota-rollout-engine, ota-update-engine-bridge, ota-android-agent, ota-telemetry-schema)	Reuse	COVERED — master §10 new-repo table; ota-protocol

4.14 success_factors (draft §14)

ReqID	Requirement (source)	Category	Coverage	Conflict?	Folding action
R02-14.1	Scalability 10,000+ concurrent devices; single-board→millions (§14.1 L1958, §3 L18)	NFR	PARTIAL / UNVERIFIED — guarantee stated (master §1); jitter design serves it (integration_guide §6); no load proof	none	Validate via load tests (deferred)
R02-14.2	Reliability 99.9% uptime SLA (§14.1 L1959)	NFR	MISSING / UNVERIFIED — no SLA bound or HA spec	none	Out of MVP scope; future SRE spec
R02-14.3	< 100 ms API response time (§14.1 L1961)	NFR	MISSING / UNVERIFIED — no perf budget defined	none	Defer; set from load tests
R02-14.4	Compatibility: Android 15 (API 35+), Linux 5.10+, Windows 10+ (§14.1 L1962)	NFR	PARTIAL — Android 15 COVERED; Linux/Windows future	none	Future phases
R02-14.5	Quality gates: zero critical vulns, security scanning, integration testing (§14.2 L1964–1970)	NFR/ Quality	PARTIAL — four-layer testing COVERED; CI security scanning not specified	none	Add to CI spec
R02-14.6	Risk mitigation: abstract hawkBit behind interface; horizontal scaling; DR/ backups (§14.3 L1972–1978)	NFR	PARTIAL — hawkBit abstraction COVERED (ADR-0001 gated behind interface); DR/backup/ sharding not specified	K4	DR/ backup is future ops spec

5. unverified_register

Per §7.1 / §11.4.6 (UNVERIFIED clause numbers), claims that could not be confirmed against an authoritative source:

- **HelixConstitution clause numbers** (§1, §7.1, §11.4.6, §11.4.8, §11.4.28, §11.4.61, §11.4.74) — carried from corpus convention; the Constitution text is not in this repository.
- **hawkBit GitHub stats** (R02-1.2) and **“Mender client is C++”** (R02-1.3) — draft-02 specifics, not independently re-verified; not propagated as fact.
- **Android-15 optional payload_properties headers** (SWITCH_SLOT_ON_REBOOT, RUN_POST_INSTALL, DISABLE_DOWNLOAD_RESUME) (R02-2.6) — UNVERIFIED against AOSP 15 (carried from aosp-update-engine open items).
- **Catalogue-brick public surfaces** (auth, security, Storage, database, http3, ratelimiter, middleware) — reuse claims are conditional on inspection (submodule_reuse_map Continuation).
- **NFR numbers** (10,000+ devices, 99.9% uptime, <100 ms) (R02-14.x) — asserted in draft, not measured or bound anywhere in the corpus.
- **Range-over-HTTP/3** and **Brotli quality tuning** — UNVERIFIED pending the ADR-0004 §6 spike.

6. sources

- additions/initial_research_02.md (the analyzed draft; line cites above).
- research/additions_synthesis.md — Rev 1 high-level synthesis (this doc is its granular successor for draft 02).
- 00-master/2026-06-07-helix-ota-design.md — master design (§1 guarantees/non-goals, §2 locked decisions, §3 stack, §4 architecture/OS-adaptor seam, §5 MVP, §6 trust, §7 data model, §8 rollout, §9 telemetry, §10 reuse, §11 phasing, §13 testing).
- research/adr/adr-0001-wrapped-engine.md ... adr-0005-delta-updates.md — conflict resolutions K1–K5.
- 1.0.0-mvp/api/endpoints.md, database/schema.md, security/signing_verification.md, security/key_management.md, client_android/integration_guide.md, client_android/update_engine_integration.md, server/artifact_validation.md, server/telemetry_processing.md, deployment/overview.md, tests/test_strategy.md.
- 1.0.1-staged-rollout/README.md; 1.X-linux/README.md, 1.X-windows/README.md, 1.X-other-os/README.md.
- research/stacks/* — eclipse-hawkbit, mender, tuf-go-tuf, uptane, rauc, swupdate, android15-virtual-ab, aosp-update-engine, android-update-engine-api, android-avb-rollback.
- Built server: server/internal/api/server.go (routes), wire.go (wire types), handlers_*.go, middleware.go, token.go, users.go; server/internal/store/store.go (Repository).